Coordinated Justice: The Politics of Prosecution and Policing[*]

Tom S. Clark[†]

August 5, 2026

Abstract

Criminal justice outcomes emerge from the coordinated actions of police and prosecutors, yet these institutions are typically studied in isolation. We lack a theoretical framework for understanding how the strategic incentives of police agencies and elected prosecutors interact to jointly determine enforcement patterns. This paper develops a game-theoretic model in which police choose evidence investment and enforcement allocation while prosecutors set evidentiary standards for charging. The analysis reveals a strategic complementarity between these choices that generates multiple equilibria and institutional path dependence in enforcement regimes. Identical formal policies can sustain sharply different enforcement environments depending on coordination history, and reforms that fail to account for this interdependence may produce null or heterogeneous effects. The framework yields direct implications for the empirical analysis of criminal justice reform, offering guidance for interpreting estimated reform effects.

[*]I thank Steve Callander, Sam England, David Fortunato, Scott Gehlbach, and Dan Kessler for helpful comments and discussion.

[†]Professor of Political Science, Stanford University, tsclark@stanford.edu

When Larry Krasner took office as Philadelphia's District Attorney in 2018, he changed which cases his office would pursue. He instructed his attorneys to decline a range of low-level charges, to divert others, and to treat most low-level retail theft as a summary offense. These were choices about which kinds of offenses to prioritize, not about how much evidence a case needed to contain before it could be filed. Over the ensuing years the changes generated sustained controversy and tension with the Philadelphia Police Department. Yet despite the shift in charging priorities, citywide arrest and conviction rates did not exhibit a clear structural break. In March 2025, Krasner's office partially reversed course, and in November 2025 he was re-elected to a third term.

A similar agenda unfolded in San Francisco. When Chesa Boudin took office as District Attorney in January 2020, he too narrowed the set of cases his office would pursue, declining to prosecute quality-of-life offenses such as public camping, solicitation, and public urination. As in Philadelphia, the changes were defined by the type of offense rather than by the evidence in any particular case, and they provoked backlash from police officials. Unlike in Philadelphia, they coincided with visible changes in enforcement patterns and became a focal point of political contestation. In 2022, Boudin was removed from office in a recall election.

These cases present a puzzle. Two prosecutors in two large cities adopted broadly similar reforms, yet the system-level consequences diverged sharply. It is tempting to conclude that one reform succeeded and the other failed. I argue instead that the reforms differed less in their content than in how police and prosecutors had been coordinating beforehand, and that this difference has direct consequences for what we can infer from the data. Existing empirical designs treat changes in charging or conviction rates as direct reflections of prosecutorial policy. But if police adjust their behavior in anticipation of how prosecutors will treat cases, those rates conflate the selection of cases into the prosecutorial pipeline with outcomes conditional on entry, and standard designs may misattribute a reform's effects or miss them entirely. Criminal justice enforcement is the joint product of strategic interaction between prosecutors and police, and much of the politics of criminal justice resides in that interaction.

I develop a theory of criminal justice enforcement as a coordination problem between police and prosecutors. Police identify suspects, collect evidence, and assemble case files. Prosecutors decide whether to file charges and what charges to bring. Prosecutors influence enforcement through more than one instrument. The Krasner and Boudin reforms operated on offense priorities, the set of offenses an office chooses to pursue. A second instrument, less visible but also consequential, is the evidentiary standard that governs how much evidence a case must contain before the office will file it. Offices set this standard through internal screening rules such as corroboration requirements, supervisory approval, or body-camera compliance. This paper focuses on the evidentiary standard and on how it interacts with the mix of offenses that police bring forward. Police allocate effort across areas of law enforcement, and invest in building case files, in anticipation of how prosecutors will treat what they produce. Prosecutors, especially elected ones, set their standards in the shadow of electoral accountability. Prosecutorial standards and police evidence investment are therefore strategic complements, and criminal justice enforcement has the structure of a coordination game. When two choices are strategic complements, the system they define can settle into more than one stable configuration. Path dependence, equilibrium multiplicity, and reform fragility are then properties of the institutional system itself, not artifacts of measurement or implementation failure.

This interaction has not gone unnoticed. Empirical work on prosecutorial reform increasingly asks whether police respond to changes in charging policy, and it often treats that response as a threat to identification to be detected and controlled (e.g., Agan, Doleac and Harvey 2023, Ouss and Stevenson 2023). Scholars of policing and prosecution in criminology, law, and sociology have long argued that the two functions are deeply interdependent (e.g., Richman 2003, Sklansky 2018). What this work has lacked is a framework that characterizes the structure of the interdependence: when police and prosecutors coordinate on one enforcement pattern rather than another, why similar reforms can produce different outcomes, and what those outcomes imply for inference. The prosecutorial-discretion literature has largely studied charging while taking the pool of cases as given (e.g., Gordon and Huber 2002, Landes 1971), and the policing literature has largely studied enforcement while taking downstream charging standards as fixed (e.g., Wilson 1968). Studied jointly, each margin turns out to depend on the other in ways that matter for what we can infer from aggregate outcomes such as arrest, charging, and conviction rates (e.g., Agan et al. 2025).

I formalize the interaction in a model in which a prosecutor sets an evidentiary standard for filing charges and police respond by allocating effort across areas of criminal law and investing in evidence production. Three results follow. First, higher charging standards need not reduce conviction rates. They operate on the selection margin and change the composition of the pool of prosecuted cases, which runs against the intuition that tighter standards must produce fewer convictions.

Second, prosecutorial standards shape police effort and priorities, and they can generate multiple enforcement equilibria that differ in both evidence production and charging behavior. Identical formal policies can then sustain different enforcement environments, depending on how police and prosecutors coordinate. Third, a reform that does not alter the underlying coordination may produce no observable change in aggregate outcomes even when it meaningfully changes formal policy. That stability reflects equilibrium persistence in a system of strategic complements, not the irrelevance of prosecutorial discretion.

The framework speaks to a growing empirical literature on the effects of prosecutorial reform on case outcomes (e.g., Agan, Doleac and Harvey 2023, Agan et al. 2025, Bandyopadhyay and McCannon 2014, Dobbie, Goldin and Yang 2018, Goncalves and Mello Forthcoming, Pfaff 2017, Soliman 2022) and on police behavior (e.g., Ba et al. 2021, Clark 2024, Eckhouse 2022, Mummolo 2018). Its contribution is not the observation that policing and prosecution are interdependent, which is well established, but a formal account of how the two are jointly determined and what that joint determination implies. Treating criminal justice as an equilibrium outcome of coordinated behavior clarifies why empirical evaluations may yield null or heterogeneous effects, identifies observable regime signatures that distinguish enforcement equilibria, and shows that the tools political scientists use to evaluate reform must account for the coordination dynamics that produce them.

1 Prosecutors and Police

Recent literature has developed two increasingly sophisticated but largely separate research programs on the central actors in American criminal law enforcement. One studies prosecutors: their unilateral charging authority, the role of electoral accountability in shaping their behavior, and the consequences of their discretion for who is prosecuted and on what terms. The other studies police: their organizational incentives, resource allocation decisions, and the determinants of enforcement patterns. Each tradition has generated important insights, but each also takes as given features of the institutional environment that are, in fact, endogenous to the other actor's choices. The result is that our theoretical understanding of criminal justice politics is built on a foundation that treats as separate what is, in practice, a single interdependent system. By operating largely in isolation, these literatures overlook important dynamics that complicate their inferences.

Prosecutors exercise unilateral formal authority over charging decisions and are directly accountable to the public; police are far less electorally accountable. The literature on prosecutorial discretion emphasizes the consequences of that discretion for who is prosecuted and on what terms (e.g., Barkow 2019, Clark Forthcoming, Davis 2007, Gordon and Huber 2009, Landes 1971), and treats electoral accountability as the prosecutor's primary constraint (e.g., Gordon and Huber 2002). But these analyses typically model the prosecutor as facing an exogenous distribution of cases. That assumption is untenable once we recognize that police anticipate and respond to prosecutorial standards: the pool of cases a prosecutor faces is itself a strategic object, shaped by the policies the prosecutor adopts.

Prosecutors depend heavily on police, who determine which cases are investigated, what evidence is collected, and how cases are packaged for prosecution (Goldstein 1963, Greenwood, Chaiken and Petersilia 1977, Lee 2020). Prosecutors, unlike police, are politically accountable for the state of public safety. That accountability is usually described in electoral terms, but most chief prosecutors run unopposed, and local electoral competition is, on its own, weak (Hessick and Morse 2019, Rogers 2023, Wright 2009). What the model requires is not contested elections but political accountability broadly. Reputational pressure, media scrutiny, career concerns, and the latent threat of a challenge can each discipline charging policy. I therefore treat elections as the most visible instance of a more general accountability pressure, and nothing in the analysis turns on that pressure operating through close races.

The literature on police behavior and organizational incentives largely treats downstream charging standards as fixed rather than as strategic choices that reshape police incentives (e.g., Goldstein 1963, Goncalves and Mello Forthcoming, Soliman 2022, Wilson 1968). But an agency anticipating strict prosecutorial screening finds it costly to pursue cases without the requisite evidentiary foundation, and may redirect effort toward categories where prosecutable evidence can reliably be produced. When prosecutors adopt permissive standards, the marginal return to thorough investigation declines, and police may instead prioritize arrest volume over case quality (Forst and Brosi 1977, Pfaff 2017). Prosecutorial standards thus function as a form of institutional delegation, shaping the composition and quality of cases that enter the system without prosecutors exerting direct authority over policing (Sklansky 2018).

This arrangement separates formal authority from practical influence: prosecutors control charging, but police control the informational inputs on which charging rests (e.g., Richman 2003). The interaction takes place in the shadow of a court. The risk of acquittal and the reputational cost of failed prosecutions constrain the cases a prosecutor can viably charge, making the quality of police-generated evidence a matter of political salience rather than administrative preference (Bandyopadhyay and McCannon 2014, Bibas 2004, Boylan and Long 2005, Eisenstein and Jacob 1977, Nardulli, Eisenstein and Flemming 1988). The constraint propagates backwards: prosecutors screen with an eye toward trial viability, and police, anticipating that screening, produce evidence that meets the prosecutor's threshold (Albonetti 1987, Forst and Brosi 1977). The court's shadow therefore propagates through two strategic actors, and the standard principal-agent framework, with its exogenous case pool, cannot capture it. Political pressure on prosecutors need not translate mechanically into charging outcomes: because police reallocate effort in anticipation of screening, the mapping from public preferences to enforcement is mediated at every stage by the interaction between the two actors (Gordon and Huber 2002, Huber and Gordon 2004, Wilson 1968).

Recent work has begun to relax these assumptions, treating the police response to prosecutorial policy as an object of study rather than a maintained assumption (e.g., Agan, Doleac and Harvey 2023, Ouss and Stevenson 2023). What remains missing is an account of the joint determination of prosecution and policing: how the strategic interaction between an electorally accountable prosecutor and a law enforcement agency shapes what laws get enforced, who is prosecuted, and how well the system distinguishes strong cases from weak ones.

The gap is sharpest in formal work. Formal models of prosecution have largely taken the prosecutor's case pool as given and focused on optimization within it, including investigative resource allocation and plea bargaining (Baker and Mezzetti 2001), charging under case-level uncertainty (Bjerk 2007), and the objectives prosecutors pursue in selecting among federal cases (Glaeser, Kessler and Morrison Piehl 2000). Related work on optimal law enforcement (Shavell 1993) models enforcement intensity but not the strategic production of evidence by a separate agency with its own accountability. The present model treats prosecution and policing as a joint coordination problem, in which equilibrium multiplicity and institutional path dependence arise from the strategic complementarity between an electorally accountable principal and a semi-autonomous evidence-producing agency.

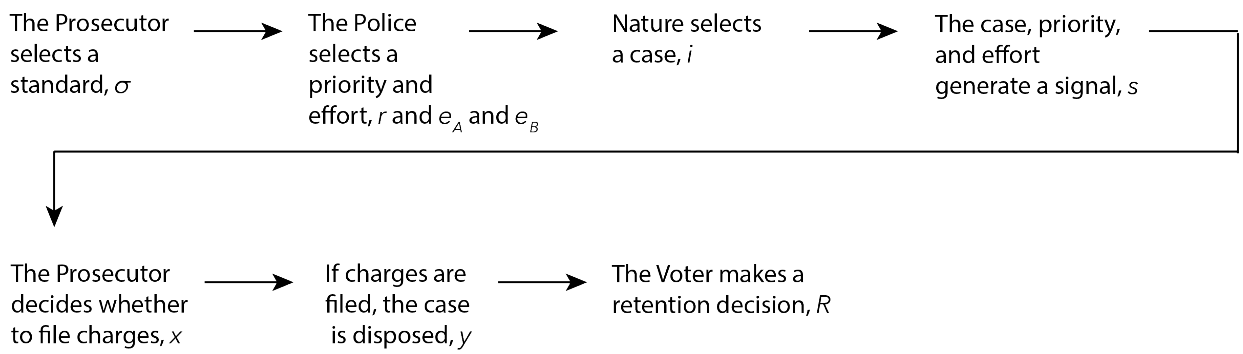

Figure 1: Timing of events in the model

2 The Model

I model an institutional setting in which a police agency produces evidence and an elected prosecutor exercises charging discretion under electoral accountability. The model isolates two strategic channels: (i) how police-produced information affects prosecutorial charging decisions, and (ii) how prosecutorial evidentiary standards shape police priorities and investment incentives across different areas of law enforcement. For expositional convenience, I describe the model as a single-case game. Conceptually, however, the game is played repeatedly across a large number of cases, and the analysis below treats equilibrium behavior as generating outcomes over a portfolio of cases.

2.1 Sequence of Play

There are two strategic players: an elected prosecutor P and a police agency L. The agency acts through a commissioner, who sets enforcement priorities, and line officers, who choose investigative effort; the two share the agency's objective, so L is a single player that moves in two stages rather than two players with competing interests. Two non-strategic actors, a trial court J and a voter V, complete the model. The game entails seven steps, as follows.

1. **Prosecutorial policy choice.** The prosecutor commits to an evidentiary standard $\sigma \in \{\text{strict}, \text{lenient}\}$, publicly observed by the agency. Formally, σ is a commitment to a *feasible set of admissible charging rules* $\mathcal{X}_\sigma$ (specified at the charging stage below), not to a single rule. This distinction is what allows a strict standard to fully determine charging while a lenient standard leaves the prosecutor residual discretion.

2. **Police priorities.** Observing σ, the police commissioner chooses a priority parameter $r \in$

$[0, 1]$ that determines the distribution of cases across activity types $i \in \{A, B\}$, with $\Pr(i = A) = r$.

3. **Police effort.** Observing (σ, r), line officers choose evidence investments $e_A, e_B \in [0, \bar{e}]$, subject to a total effort budget

$$r\, e_A + (1 - r)\, e_B \ \le\ \bar{B}.$$

Because effort is chosen after the allocation, the commissioner anticipates the officer response when setting r.

4. **Case emergence and signal generation.** Nature draws a representative case, $i \in \{A, B\}$, where $\Pr(i = A|r) = r$ and $\Pr(i = B|r) = 1 - r$, with an unobserved underlying state $\theta \in \{0, 1\}$. Conditional on i, $\Pr(\theta = 1 \mid i) = \pi_i$, where $\theta = 1$ denotes a guilty defendant and $\theta = 0$ an innocent one. Whether a guilty defendant is ultimately convicted is endogenous, determined below by the evidence and the court. Conditional on (θ, i, e_i), the case file produces a binary signal $s \in \{H, L\}$ representing evidentiary strength:

$$\Pr(s = H \mid \theta = 1, i, e_i) = \alpha_i(e_i), \qquad \Pr(s = H \mid \theta = 0, i, e_i) = \beta_i(e_i),$$

I assume $\alpha_i'(e) > 0$ and $\beta_i'(e) \ge 0$ and that $\alpha_i'(e) > \beta_i'(e)$. That is, the more effort the police agency invests, the greater the likelihood they produce a strong file, and their effort increases that probability more when the defendant is guilty. The prosecutor observes (i, s) but does not observe θ or e_i.

5. **Charging.** After observing (i, s), the prosecutor selects a sequentially rational charging rule $x : \{A, B\} \times \{H, L\} \to \{0, 1\}$ from the admissible set $\mathcal{X}_\sigma$ fixed by her standard, where

$$\mathcal{X}_{\text{strict}} = \{x :\ x(i, H) = 1,\ x(i, L) = 0\}, \qquad \mathcal{X}_{\text{lenient}} = \{x :\ x(i, H) = 1,\ x(i, L) \in \{0, 1\}\}.$$

Under a strict standard the admissible set pins down the weak-signal margin, so charging is fully determined. Under a lenient standard the set leaves the weak-signal margin free, and the prosecutor's choice within $\mathcal{X}_{\text{lenient}}$ is her residual discretion—the object whose interaction with police effort generates the multiplicity analyzed below.

6. **Adjudication.** If $x = 1$, the case proceeds to court and produces outcome $y \in \{0, 1\}$, where $y = 1$ denotes conviction. Conviction probability depends on the underlying state and the signal:

$$\Pr(y = 1 \mid x = 1, \theta, s) = q(\theta, s),$$

   with conviction more likely when the defendant is guilty and the evidentiary signal is high.[1] I assume the shape of $q()$ is exogenous and reflects the judicial environment in which a case is prosecuted. If $x = 0$, no adjudication occurs.

7. **Elections.** The voter observes coarse, aggregate summary measures of prosecutorial performance generated by the portfolio of cases handled during the period (e.g., convictions, failed prosecutions, and declinations), as well as the prosecutor's announced evidentiary standard σ. Based on these summaries, the voter decides whether to retain the prosecutor, $D \in \{0, 1\}$, with $D = 1$ denoting the decision to retain the prosecutor. Below, I formalize the mapping from aggregate performance to retention.

Table 1 summarizes the model parameters and choices and the roles they play in the game, and Figure 1 summarizes the timing.

2.2 Elections and Aggregation

Although the sequence of play is written at the level of a single case, the model should be interpreted as describing a representative case drawn from a larger portfolio of cases handled over an election period. The Voter does not condition reelection on any one case outcome in isolation. Instead, she observes coarse summary measures of office performance (e.g., charging rates, conviction rates, and salient failures) generated by the portfolio.

Let $\Gamma(\cdot)$ map aggregate summary measures of performance into the prosecutor's probability of retention. For expositional convenience, I write Γ as a function of convictions, acquittals, and declinations, though the formal objects are the corresponding election-period rates (or their expec-

[1] Formally, we might just assume that q is increasing in θ and in s, but of course each of those is dichotomous.

Parameter	Definition
$\sigma \in \{\text{strict}, \text{lenient}\}$	The prosecutor's *evidentiary standard*. This is an ex ante policy choice that constrains how signals are translated into charging decisions.
$r \in [0, 1]$	The police agency *priority choice*. It determines the distribution of cases across activity types $i \in \{A, B\}$.
$e_i \in [0, \bar{e}]$	Police agency *evidence investment*. It affects the quality and diagnostic value of the case file.
$\theta \in \{0, 1\}$	The *underlying legal state*. $\theta = 1$ denotes a guilty defendant and $\theta = 0$ an innocent one. Guilt is never directly observed; conviction depends on it through the evidence and the court.
$s \in \{H, L\}$	The *evidentiary signal* contained in the file. This is what the prosecutor observes about the case.
$x \in \{0, 1\}$	The prosecutor's *charging decision*. $x = 1$ denotes filing charges; $x = 0$ denotes declination.
$y \in \{0, 1\}$	The *adjudication outcome*. Conditional on $x = 1$, $y = 1$ denotes conviction and $y = 0$ denotes failure to convict.

Table 1: Key model objects and their roles

tations). Specifically, let

$$\bar{y} \equiv \mathbb{E}[y], \qquad \overline{x(1-y)} \equiv \mathbb{E}[x(1-y)], \qquad \overline{(1-x)} \equiv \mathbb{E}[1-x],$$

where the expectation is taken over the election-period distribution of cases (induced by r and equilibrium evidence investment). The prosecutor's retention probability is then

$$\Pr(D = 1 \mid \text{history}) = \Gamma\Big(\bar{y},\ \overline{x(1-y)},\ \overline{(1-x)},\ \sigma\Big).$$

This interpretation preserves the single-case decision problem while capturing the fact that electoral accountability is generated by performance aggregated across many cases, including cases not prosecuted (declinations). Also, I impose minimal structure on $\Gamma(\cdot)$, allowing voters to evaluate performance using either conditional outcomes (e.g., conviction rates among prosecuted cases) or unconditional outcomes (e.g., conviction or declination rates across all cases). The results below do not depend on this distinction, and hold for a broad class of evaluation rules that reward successful prosecutions and penalize failed or unjustified cases. Because Γ enters only in reduced form, elections operate here as a device that disciplines the prosecutor's choices, not as a mechanism for selecting among prosecutor types. This also bounds the role of voter sophistication. A strategic voter who updated on the composition of outcomes would induce some evaluation rule, and because the results hold for a broad class of such rules, endogenizing the voter would not overturn the equilibrium behavior characterized below.

2.3 Preferences

Prosecutor. I assume the prosecutor values convictions, dislikes losing charged cases, dislikes convicting the innocent, and values holding office. Normalize the per-case payoff as

$$U_P \;=\; b \cdot y \;-\; a \cdot x(1-y) \;-\; \lambda \cdot y(1-\theta) \;+\; \rho \cdot \Pr(D=1 \mid \text{history}),$$

where $b > 0$ captures the professional and reputational benefit of a conviction, $a > 0$ the reputational cost of a failed prosecution (especially a salient acquittal), $\lambda > 0$ the normative cost of convicting an innocent defendant (the event $y = 1$ while $\theta = 0$), and $\rho > 0$ the value of retention. The intrinsic terms b and a capture stakes—professional standing, office morale, credibility with police and the courts—that the coarse, binary retention outcome does not fully register; electoral accountability operates separately through Γ. The normative cost attaches to the outcome, a wrongful conviction, rather than to the act of charging. I normalize the payoff from declining to charge to 0.

Police agency. The police agency values having cases charged (and, optionally, convicted) and incurs costs of (i) allocating enforcement attention across categories and (ii) producing evidentiary files. The agency values charges and convictions as enforcement output rather than conditioning on guilt, consistent with the prominence of arrest and clearance statistics as measures of police productivity. Let $r \in [0,1]$ denote the induced share of cases in category A. The police agency's expected payoff is

$$\begin{aligned}\mathbb{E}[U_L] \;&= r\Big(v_A \Pr(x=1 \mid A) + w_A \Pr(y=1 \mid A) - c(e_A)\Big) \\ &\quad +\; (1-r)\Big(v_B \Pr(x=1 \mid B) + w_B \Pr(y=1 \mid B) - c(e_B)\Big) \;-\; C(r),\end{aligned}$$

where c is increasing and convex and C is convex with an interior minimum. Here $c(e_i)$ is the per-case resource cost of generating evidentiary quality in category i. $C(r)$ measures the organizational cost of *concentrating* enforcement—shifting patrol assignments, standing up or disbanding detective units, and the political frictions of reallocation—and is minimized at an interior allocation, so that concentrating resources in either category is costly.

The agency is not a single actor. The commissioner chooses the allocation r to maximize $\mathbb{E}[U_L]$, anticipating officer effort; line officers then choose (e_A, e_B) to maximize $\mathbb{E}[U_L]$ given r, subject to the total effort budget $re_A + (1-r)e_B \leq \bar{B}$. The two frictions play distinct roles. The effort budget couples the allocation and effort choices: because investigative capacity is shared, the effort officers can devote to a case depends on how many cases the commissioner routes into each category. The reallocation cost $C(r)$ keeps the allocation interior. Both are operative, and each governs a different margin.

2.4 From representative cases to aggregate outcomes

Although the model is written as a representative-case game, the objects of empirical interest are aggregates computed over a large portfolio of cases within an election period. It is therefore useful to make explicit how individual case realizations map into observable quantities. A *case draw* corresponds to a single realization of (i, θ, s) induced by agency choices (r, e_A, e_B). The priority parameter r determines the distribution of case types, with $\Pr(i = A) = r$ and $\Pr(i = B) = 1 - r$, while evidence investments e_i determine the signal distribution via $\alpha_i(e_i)$ and $\beta_i(e_i)$. The joint distribution of (i, s, θ) therefore depends endogenously on agency behavior.

Observed outcomes are frequencies over this induced distribution. Let

$$\bar{x} \equiv \Pr(x = 1), \quad \bar{y} \equiv \Pr(y = 1), \quad \bar{d} \equiv \Pr(x = 0)$$

denote the charging rate, unconditional conviction rate, and declination rate, respectively, where probabilities are taken over the equilibrium distribution of cases. The conditional conviction rate is $\Pr(y = 1 \mid x = 1)$. Because police agency choices determine the distribution of cases that enter the system, they also affect the denominators underlying these quantities. In particular, the set of "potential cases" is itself endogenous: changes in r shift the composition of cases across categories, while changes in e_i affect the likelihood that cases generate high- or low-quality signals. As a result, both the numerator and denominator of observed rates reflect equilibrium behavior.

Changes in observed charging or conviction rates may therefore arise from shifts in prosecutorial decisions, changes in the composition of cases induced by police behavior, or both, a point that shapes the empirical implications developed below.

2.5 Notable modeling choices

Several set-up choices are defended in Appendix A-2: the binary (rather than continuous) evidentiary standard σ; the prosecutor's inability to observe police effort e_i directly; the reduced-form court $q(\cdot)$; the treatment of the police agency as two roles, a commissioner who sets priorities and line officers who choose effort under a shared budget; the scope of the weak-file margin to contested cases; and the timing of evidence production relative to charging. The most consequential is the binary σ. Real offices adopt coarse, verifiable filing policies rather than continuous thresholds, and the strategic complementarity at the model's core does not depend on admitting more than two values of σ.

3 Analysis

To analyze the model, I first describe the core logic underlying the prosecutor and police agency's decision calculus. I then turn to an equilibrium analysis and the behavioral properties that arise in equilibrium. Section 4 then builds on this characterization to examine coordination, equilibrium selection, and the implications of multiplicity for reform.

3.1 The Prosecutor's Charging Condition

Although the prosecutor's evidentiary standard σ is chosen ex ante as a commitment device, not every standard is compatible with the prosecutor's incentives given the police behavior it induces. Let the prosecutor's *charging compatibility condition* be the incentive condition under which a given evidentiary standard can be sustained in equilibrium. The condition does not describe discretionary case-by-case behavior in isolation. Rather, it characterizes how prosecutorial preferences and electoral incentives jointly discipline both charging decisions and the equilibrium police investment that the prosecutor rationally anticipates under a chosen standard.

To that end, let $\mu_i(s;\sigma)$ denote the prosecutor's posterior belief that a defendant from activity type i is guilty, given the evidentiary signal s and the prosecutor's chosen standard σ. Formally,

$$\mu_i(s;\sigma) \equiv \Pr(\theta = 1 \mid i, s, \sigma) = \frac{\pi_i \Pr(s \mid \theta = 1, i, \sigma)}{\pi_i \Pr(s \mid \theta = 1, i, \sigma) + (1 - \pi_i) \Pr(s \mid \theta = 0, i, \sigma)},$$

where $\pi_i \equiv \Pr(\theta = 1 \mid i)$ is the prior probability that a case of type i involves a guilty defendant. Note that because the evidentiary standard σ is publicly observed and shapes police agency incentives, it pins down an equilibrium level of police evidence investment $e_i(\sigma)$, which is not directly observed by the prosecutor. The prosecutor therefore conditions her charging decision on σ, and the induced distribution of signals.

Next, define the per-case *electoral marginal incentive* from charging as

$$\Delta_E(i, s; \sigma) \equiv \rho \left(\mathbb{E}[\Gamma(\cdot) \mid x = 1, i, s, \sigma] - \mathbb{E}[\Gamma(\cdot) \mid x = 0, i, s, \sigma] \right).$$

This object captures how the electoral environment rewards or penalizes the act of filing charges, conditional on the evidentiary class of the case and the prosecutor's announced standard. That is, it measures how much more electoral return the prosecutor gets from charging a given case, as opposed to not charging it.

With these objects in hand, the prosecutor's charging condition is direct. For any (i, s) for which charging is permitted by the chosen standard σ, the prosecutor files charges if and only if the expected payoff from charging weakly exceeds the payoff from declination:

$$\begin{aligned} x(i, s; \sigma) = 1 \quad \iff \quad & b \cdot \Pr(y = 1 \mid x = 1, i, s) - a \cdot \Pr(y = 0 \mid x = 1, i, s) \\ & - \lambda \cdot \Pr(\theta = 0 \mid i, s, \sigma)\, q(0, s) + \Delta_E(i, s; \sigma) \geq 0, \end{aligned}$$

where

$$\Pr(y = 1 \mid x = 1, i, s) = \mu_i(s;\sigma)\, q(1, s) + (1 - \mu_i(s;\sigma)) q(0, s).$$

This charging condition reveals the central incentive tradeoff at the heart of the model. The prosecutor weighs the substantive and electoral benefits of a potential conviction against three countervailing forces: the reputational cost of an acquittal, the normative cost of convicting an innocent defendant, and the electoral consequences of filing charges. Importantly, because beliefs about case quality depend on the police response induced by σ, the condition simultaneously determines (a) which evidentiary standards can be sustained in equilibrium and (b) how police investment must adjust to make those standards incentive compatible. As I show in the equilibrium analysis below, these constraints play a central role in shaping police priorities and the existence of distinct equilibrium charging regimes.

3.2 Police priority allocation and effort investments

Consider next the police agency's incentives, solving the two stages by backward induction: line officers choose effort given the allocation, and the commissioner then chooses the allocation anticipating that effort. To see how these forces interact, let

$$\Pi_i(\sigma) \;\equiv\; v_i \Pr(x{=}1 \mid i, \sigma) \;+\; w_i \Pr(y{=}1 \mid i, \sigma)$$

denote the police agency's expected per-case gross return from a case in category i under standard σ, evaluated at the equilibrium effort induced by σ.

Officer effort. Given (σ, r), officers choose (e_A, e_B) to maximize the agency payoff subject to the effort budget. Letting $\mu \geq 0$ be the multiplier on $re_A + (1-r)e_B \leq \bar{B}$, the interior conditions are

$$v_i \frac{\partial \Pr(x{=}1 \mid i)}{\partial e_i} \;+\; w_i \frac{\partial \Pr(y{=}1 \mid i)}{\partial e_i} \;-\; c'(e_i) \;=\; \mu, \qquad i \in \{A, B\},$$

with complementary slackness $\mu\,(re_A + (1-r)e_B - \bar{B}) \;=\; 0$. Officers set effort so that the net marginal return to investigation equals the common shadow price μ of investigative capacity. When the budget is slack, $\mu = 0$ and each category's marginal return is equated to its marginal cost; when it binds, effort is pulled down until net marginal returns equal μ, and effort in each category depends on the allocation r through the binding budget.

The two terms on the left are the channels through which evidence quality generates value for the police: raising the probability that the prosecutor files charges (weighted by v_i) and the probability of eventual conviction (weighted by w_i). Their relative importance depends on σ.

Under a strict standard, charging is sensitive to evidentiary strength, so both terms are active and investigation carries a dual return. Under a lenient standard, the prosecutor charges regardless of signal quality, so $\Pr(x{=}1 \mid i)$ is approximately invariant to e_i, the first term vanishes, and the only return to investigation is through conviction. Because strict screening activates the charging channel, it raises the demand for effort; the budget is then more likely to bind, so the shadow value of capacity tends to be higher under strict screening.[2]

Commissioner priorities. Anticipating officer effort $e_i(r;\sigma)$ and the shadow price $\mu(r;\sigma)$, the commissioner chooses r to solve $\max_{r\in[0,1]}\ r[\Pi_A(\sigma) - c(e_A)] + (1-r)[\Pi_B(\sigma) - c(e_B)] - C(r)$. By the envelope theorem, an interior solution satisfies

$$[\Pi_A(\sigma) - c(e_A)] - [\Pi_B(\sigma) - c(e_B)] - \mu\,(e_A - e_B) \;=\; C'(r), \tag{1}$$

which implicitly defines the equilibrium priority choice $r^*(\sigma)$. The term $-\mu(e_A - e_B)$ is an opportunity cost: routing one more case into a category draws officer effort into that category, and when investigative capacity is scarce that reallocation is priced at its shadow value μ. The prosecutor's standard therefore affects police behavior along two margins. It shapes effort *within* categories, and it shifts priorities *across* categories, both directly through the relative returns $\Pi_A - \Pi_B$ and indirectly through the shadow price μ. (These observations are central to several results below on the empirical implications of prosecutorial reform.)

This distinction drives the conditions that support distinct types of equilibrium behavior in the following analysis: the value of investigative effort to the police agency depends on whether downstream charging is responsive to evidence, which in turn depends on the prosecutor's choice of screening standard. The analysis I present next characterizes how this strategic interdependence between prosecutorial charging policy and police evidence production gives rise to equilibrium behavior.

[2] This is a tendency rather than a general inequality; it holds throughout numerical exploration of the model but a formal proof requires a monotone-comparative-statics argument.

3.3 Equilibrium Regimes

To solve the model, I characterize pure-strategy Perfect Bayesian Equilibria (PBE). A PBE comprises (i) a prosecutor policy choice σ and charging rule $x(i, s; \sigma)$, (ii) police agency choices r and $e_i(\sigma)$, and (iii) posterior beliefs, such that:

(a) the prosecutor's choice of σ and subsequent charging rule $x(\cdot)$ maximize $\mathbb{E}[U_P]$ given police strategies and Γ;

(b) the police agency choices of r and $\{e_i\}$ maximize $\mathbb{E}[U_L]$ given σ and the induced charging rule; and

(c) posterior beliefs $\mu_i(s; \sigma)$ are derived via Bayes' Rule wherever applicable.

Because Γ is a reduced-form function of aggregate case outcomes, the voter does not engage in strategic optimization. The equilibrium is therefore fully characterized by the strategies of the prosecutor and police agency together with the prosecutor's posterior beliefs about case quality.

It will be useful throughout the equilibrium analysis to define the *net marginal gain from charging* a case of type i with signal s under standard σ:

$$\Phi(i, s; \sigma) \;\equiv\; b \cdot \Pr(y{=}1 \mid x{=}1, i, s) \;-\; a \cdot \Pr(y{=}0 \mid x{=}1, i, s) \;-\; \lambda(1 - \mu_i(s; \sigma))q(0, s) \;+\; \Delta_E(i, s; \sigma). \tag{2}$$

By assumption, the prosecutor charges when the case file is strong under either standard. That assumption imposes the parameter constraint that $\Phi(i, H; \sigma) > 0$, for $i \in \{A, B\}$, which in turn imposes constraints on the parameters themselves. What equilibrium can be supported therefore depends on (i) the sign of $\Phi(i, L; \sigma)$—the net gain from charging a weak-evidence case when such charging is admissible—and (ii) the prosecutor's (*ex ante*) expectations about payoffs under each standard. For the latter, let

$$V_P(\sigma) \;\equiv\; \mathbb{E}[U_P \mid \sigma,\, r^*(\sigma),\, e^*(\sigma)]$$

denote the prosecutor's expected payoff under standard σ, evaluated at the police agency's best response $(r^*(\sigma),\, e^*(\sigma))$. For σ to be an equilibrium strategy, then, it must be *ex ante* optimal for the prosecutor—i.e., $V_P(\sigma) \geq V_P(\sigma')$ for $\sigma' \neq \sigma$. (Recall there are only two values of σ.)

These comparisons give rise to three distinct equilibrium regimes, each arising from a distinct balance of incentives: the unscreened regime arises when electoral pressure dominates evidentiary discipline; the screened regime arises when costs of weak prosecutions dominate electoral pressure; and the category-dependent regime arises when cross-domain heterogeneity creates evidentiary discipline only in some domains. Because the analytic goal here is to connect the model to empirical work on criminal justice reform, I describe each regime in two complementary ways: by its formal structure (what prosecutors and police do in equilibrium) and by its *empirical signature*—the distinctive pattern of charging, evidence production, and case composition the regime produces in observable data. I label each regime accordingly. The unscreened regime produces *high-throughput enforcement*: charging is decoupled from evidentiary strength and enforcement tilts toward high-volume categories. The screened regime produces *evidence-driven enforcement*: charging tracks evidentiary strength and police invest in diagnostic effort. The category-dependent regime produces *domain-differentiated enforcement*: screening and evidence investment concentrate in some domains while others look unscreened. Result 1 below collects these signatures; I first lay out the equilibrium objects and, in the next section, evaluate when the system selects into one regime or another.

3.3.1 Unscreened Charging Regime (High-Throughput Enforcement)

The first regime arises when electoral incentives reward visible enforcement (i.e., charging) over evidentiary discipline (i.e., conviction). The prosecutor adopts a lenient standard and charges all cases regardless of evidentiary strength. Anticipating that evidence quality does not affect charging decisions, police underinvest in diagnostic evidence and concentrate enforcement on high-volume categories. The observable signature of this regime is therefore *high-throughput enforcement*: elevated charging rates, weak responsiveness of charging to case strength, low evidence quality, and a case mix tilted toward categories where large numbers of cases can be generated at low investigative cost.

Proposition 1 (Unscreened charging equilibrium) *Suppose that* (a) $\Phi(i, L; lenient) \geq 0$ *for all* $i \in \{A, B\}$ *and* (b) $V_P(lenient) \geq V_P(strict)$. *Then there exists a PBE in which:*

1. *the prosecutor adopts* $\sigma = lenient$*;*
2. *charging is unscreened on evidentiary strength:* $x(i, H) = x(i, L) = 1$ *for all* i*;*
3. *police priorities satisfy*

$$r^*(lenient) = \arg\max_{r \in [0,1]} \{r[\Pi_A(lenient) - c(e_A)] + (1-r)[\Pi_B(lenient) - c(e_B)] - C(r)\},$$

evaluated at the officer effort optimum under the budget (equation 1), where $\Pi_i(\text{lenient}) = v_i + w_i \Pr(y{=}1 \mid i)$*; and*

4. *police evidence investment satisfies*

$$w_i \frac{\partial \Pr(y{=}1 \mid i)}{\partial e_i} = c'(e_i) + \mu,$$

where $\mu \geq 0$ *is the shadow price of the effort budget.*

The two conditions necessary to support this equilibrium ensure that the prosecutor is willing to accept the consequences of charging weak files for both types of cases and that the prosecutor cannot do better with a strict standard, given the changes in police behavior that standard would induce. Specifically, Condition (a) requires that for each case type, the electoral benefit of charging a weak file exceeds the combined reputational cost of acquittal and normative cost of a wrongful conviction. Condition (b) requires that the prosecutor prefers the portfolio of outcomes under lenient screening—given the police behavior it induces—to those attainable under strict screening with the higher evidence investment strict would elicit.

When the prosecutor charges without screening under σ = lenient, evidence quality does not affect charging. Police priorities are therefore driven by baseline charging and conviction returns rather than by the productivity of investigative effort. Because charging does not depend on evidence quality, the police agency faces little incentive to invest in diagnostic evidence, and evidentiary signals carry limited diagnostic content. (Section 4 shows why, in the lenient regime, this low-investment environment can be self-sustaining.)

3.3.2 Screened Charging Regime (Evidence-Driven Enforcement)

A second regime arises when failed prosecutions are politically salient and wrongful charging carries meaningful costs. That incentivizes prosecutors to carefully scrutinize a file to ensure a high likelihood of conviction. The prosecutor commits to a strict standard, filing charges only when the case file is strong. The police agency responds by investing in evidence quality, concentrating effort on cases where investment can reliably produce strong files. The observable signature of this regime is *evidence-driven enforcement*: lower charging rates, strong sensitivity of charging to evidentiary strength, higher evidence quality, and a shift toward categories with high investigative productivity.

Proposition 2 (Screened charging equilibrium) *Suppose* $V_P(\text{strict}) > V_P(\text{lenient})$*. Then there* *exists a PBE in which:*

1. *the prosecutor adopts* $\sigma = strict$*;*
2. *charging is screened on evidentiary strength:* $x(i, H) = 1$ *and* $x(i, L) = 0$ *for all* i*;*
3. *police priorities satisfy*

$$r^*(strict) = \arg\max_{r \in [0,1]} \{r[\Pi_A(strict) - c(e_A)] + (1-r)[\Pi_B(strict) - c(e_B)] - C(r)\},$$

   evaluated at the officer effort optimum under the budget (equation 1), where $\Pi_i(strict) = v_i \Pr(s{=}H \mid i, e_i) + w_i \Pr(y{=}1 \mid i)$*; and*

4. *police evidence investment satisfies*

$$v_i \frac{\partial \Pr(x{=}1 \mid i)}{\partial e_i} + w_i \frac{\partial \Pr(y{=}1 \mid i)}{\partial e_i} = c'(e_i) + \mu,$$

   where $\mu \geq 0$ *is the shadow price of the effort budget.*

Strict screening introduces another incentive channel: evidence investment raises the probability that a case clears the charging threshold. The police agency therefore reallocates both investigative effort and case selection toward categories where evidence production is most effective. (Thus, as noted above, optimal effort investment includes the effect it will have on charging.) This is more likely to hold when the reputational cost of acquittals (a) is large, the wrongful-conviction cost (λ) is high, voters penalize failed cases, and the courtroom environment ($q()$) imposes meaningful scrutiny on weak evidence. That is, prosecutors face a tradeoff between inducing police effort but risking not charging cases because the agency did not produce a strong enough file, or risking losing cases because they charge cases with weak evidentiary bases.

Comparing the agency first-order conditions across regimes reveals the key incentive shift. Under strict screening, evidence investment increases the probability that a case clears the charging threshold (the v_i term), a channel that is absent under unscreened charging. The police agency therefore shifts from volume-maximizing enforcement toward diagnostic investigation. They also reallocate resources toward categories where strong evidence can be produced at relatively low cost, yielding a pattern of fewer, stronger cases.

3.3.3 Category-Dependent Regime (Domain-Differentiated Enforcement)

A final regime I consider captures selective toughness: prosecutorial screening varies across policing domains even though preferences and formal authority are uniform. This regime arises when categories differ in baseline convictability, courtroom scrutiny, or the productivity of evidence investment. The observable signature of this regime is *domain-differentiated enforcement*: heterogeneous charging thresholds across categories, with evidence quality and sensitivity to case strength improving only in domains where prosecutorial screening is effective.

Proposition 3 (Selectively screened equilibrium) *Suppose, without loss of generality, that $\Phi(A, L; lenient) \geq 0$ and $\Phi(B, L; lenient) < 0$, and that $V_P(lenient) \geq V_P(strict)$. Then there exists a PBE in which:*

1. *the prosecutor adopts $\sigma = lenient$;*
2. *charging is unscreened in category A and screened in B: $x(A, H) = x(A, L) = 1$, $x(B, H) = 1$, $x(B, L) = 0$;*
3. *police priorities satisfy*

$$r^* = \arg\max_{r \in [0,1]} \left\{ r[\Pi_A - c(e_A)] + (1-r)[\Pi_B - c(e_B)] - C(r) \right\},$$

   evaluated at the officer effort optimum under the budget (equation 1), where $\Pi_A = v_A + w_A \Pr(y{=}1 \mid A)$ and $\Pi_B = v_B \Pr(s{=}H \mid B, e_B) + w_B \Pr(y{=}1 \mid B)$; and

4. *police invest weakly in A and strongly in B, with effort in each category satisfying the officer condition (marginal net return equal to the shadow price μ).*

This regime features uniform priority allocation but differentiated investment: police tilt enforcement toward the permissive domain (where charges will be filed even in the context of a weak file) while concentrating investigative resources where prosecutorial screening rewards evidence quality. Residual discretion under σ = lenient then generates differential screening across categories. Importantly, this differential treatment does not require inconsistent preferences or discretionary favoritism; it is rational optimization in an environment where the marginal return to evidentiary discipline varies across domains.

3.4 Equilibrium Selection and Multiplicity

Unsurprisingly, under some parameter conditions, there are multiple equilibria to the game. The prosecutor's first-mover choice of σ is generically unique: she compares V_P(strict) and V_P(lenient) and selects the standard yielding the higher expected payoff. A subtler source of multiplicity, however, can arise *within* the lenient regime. Under σ = lenient, the prosecutor's commitment permits *but does not require* charging on weak files ($s = L$), and the payoff to charging weak files depends on the evidentiary environment induced by police investment. This residual discretion can therefore generate multiplicity of equilibria.

To make this interaction explicit, consider the *lenient continuation game*—the subgame that follows the prosecutor choosing $\sigma =$ lenient. Let $z \in \{0, 1\}$ denote the prosecutor's discretionary choice to charge low-signal cases, where $z = 1$ if and only if $x(i, L) = 1$, and let $e \in [0, \bar{e}]$ denote police evidence investment in the relevant category. Given an anticipated investment level e, the prosecutor chooses z by simply evaluating whether $\Phi(i, L; \text{lenient})$ is positive. Given an anticipated charging rule z, the police choose e to maximize their expected payoff net of investigative costs. An equilibrium of the lenient continuation game requires that these two best responses be mutually consistent. This is the source of equilibrium multiplicity common in coordination games.

Proposition 4 (Multiplicity under lenient screening) *Fix a category i and suppose $\sigma =$ lenient. Let $z \in \{0, 1\}$ denote the prosecutor's discretionary choice to charge low-signal cases, $z = 1 \iff x(i, L) = 1$, and let $e \in [0, \bar{e}]$ denote police evidence investment in that category. Define the prosecutor's low-signal net gain $\Phi_L(e) \equiv \Phi(i, L; \textit{lenient})$, where the posterior $\mu_i(L; \textit{lenient})$ is computed at the signal distribution induced by e. Let $BR_P(e)$ be the prosecutor's best-response correspondence for z, and let $BR_L(z)$ be the police best-response correspondence for e.*

Assume:

(i) (Monotone incentives) *$\Phi_L(e)$ is strictly decreasing in e.*

(ii) (Investment complementarity) *For any $e \in BR_L(0)$ and any $e' \in BR_L(1)$, we have $e \geq e'$, with strict inequality for interior solutions.*

(iii) (Crossing) *$\Phi_L(\min BR_L(1)) > 0$ and $\Phi_L(\max BR_L(0)) < 0$.*

Then the lenient continuation game admits at least two PBE-consistent fixed points (z, e):

1. *a* low-investment, broad-charging *fixed point with $z = 1$ and $e \in BR_L(1)$; and*
2. *a* high-investment, selective-charging *fixed point with $z = 0$ and $e \in BR_L(0)$.*

Proposition 4 isolates the coordination logic that will drive the results in the next section. In particular, it implies that identical announced standards within the lenient regime can be paired with sharply different effective screening and evidentiary environments. For a wide range of substantive settings, a lenient formal prosecutorial standard can entail either selective screening and its associated pattern of police investment and resource allocation or an unscreened regime and its associated police effort and resource allocation. In the next section, I consider the consequences: how the system selects among equilibria, why history can matter, and what that means for the empirical interpretation of reform.

4 Coordination and Institutional Path Dependence

The equilibrium regimes characterized above describe how prosecutorial screening and police evidence production interact *within* a given institutional environment. This section shows that, even holding formal standards fixed, those interactions can generate coordination problems, equilibrium multiplicity, and institutional path dependence with direct implications for reform. In particular, I show both the consequences for what we can expect from a proposed reform and how to interpret empirical patterns associated with changes in prosecutorial policies.

4.1 Comparative statics and equilibrium behavior

To begin, consider how various model parameters determine what kind of equilibrium behavior can be supported. The regimes identified above arise from the interaction of three forces: electoral incentives, the costs of weak prosecutions, and the productivity of police evidence investment. These same forces also determine whether there is a unique or multiple equilibria.

First, recall that a lenient standard (either the unscreened regime or the selective screening regime) requires $\Phi(i, L; \text{lenient}) \geq 0$, for $i \in \{A, B\}$ (unscreened) or $i = A$ (selective screening). Let

$$\mu_i^L = \frac{\pi_i(1 - \alpha_i(e_i^L))}{\pi_i(1 - \alpha_i(e_i^L)) + (1 - \pi_i)(1 - \beta_i(e_i^L))}$$

denote the Prosecutor's belief that a defendant of type i is guilty, given $s = L$. Then, the conviction probability for that case is given by

$$Q_i^L \equiv \mu_i^L q(1, L) + (1 - \mu_i^L) q(0, L).$$

Thus, the condition for $\sigma = \text{lenient}$ rearranges to

$$(b + a) Q_i^L + \Delta_E(i, L; \text{lenient}) \geq a + \lambda(1 - \mu_i^L) q(0, L), \tag{3}$$

which is a convenient expression for examining how the model's parameters determine which kinds of equilibria can be supported. The following proposition formalizes the comparative statics implied by Condition (3) and the regime characterizations in Section 3.3.

Proposition 5 (Comparative statics of regime feasibility) *Holding other primitives fixed:*

(i) The set of parameter values satisfying $\Phi(i, L; \text{lenient}) \geq 0$ is weakly increasing in the electoral marginal incentive $\Delta_E(i, L; \text{lenient})$ and in the conviction benefit b, and weakly decreasing in the acquittal cost a and the wrongful-conviction cost λ.

(ii) The set of primitives supporting the unscreened regime (Proposition 1) is therefore weakly increasing in electoral returns to visible charging and weakly decreasing in a and λ; the set of primitives supporting the screened regime (Proposition 2) has the reverse monotonicity.

(iii) The set of primitives supporting the selectively screened regime (Proposition 3) is weakly increasing in cross-category heterogeneity in the primitives $(\pi_i, \alpha_i(\cdot), \beta_i(\cdot), q(\cdot, \cdot))$ and vanishes under full symmetry across $i \in \{A, B\}$.

(iv) For fixed electoral and cost parameters, increases in the diagnostic productivity of evidence—i.e., the gap $\alpha_i(e) - \beta_i(e)$ at the equilibrium investment level—and in courtroom sensitivity $q(1, L) - q(0, L)$ raise the return to screened charging through the v_i channel in Proposition 2, weakly expanding the set of primitives supporting screened enforcement.

Proposition 5 follows directly from the definition of $\Phi(i, s; \sigma)$ in equation (2) and the regime characterizations in Propositions 1–3: each claim is a sign statement about the partial derivatives of the relevant incentive conditions. The three channels—electoral incentives, the costs of failed or unjustified prosecutions, and the productivity of evidence technology—are unpacked in Appendix A-3.

The prosecutor's first-mover choice of σ is generically unique, so multiplicity of *regimes* never arises across the strict–lenient boundary; the knife-edge $V_P(\text{strict}) = V_P(\text{lenient})$ is a measure-zero set of primitives. All genuine multiplicity is instead *intra-regime*, arising inside the lenient continuation game, where the prosecutor's residual discretion over weak files interacts with police investment. Proposition 9 in Appendix A-3 makes this precise, partitioning the primitive space into cells in which each regime is the unique equilibrium together with a distinct cell—case (ii.c)—in which the unscreened and selective regimes coexist as equilibria. That coexistence region is the subject of the next subsection.

4.2 Strategic complementarity and institutional path dependence

Case (ii.c) of Proposition 9—the region in which the crossing condition of Proposition 4 holds in the lenient continuation game—is the substantively interesting part of the primitive space. It is here that the coordination logic between prosecutors and police generates institutional path dependence,

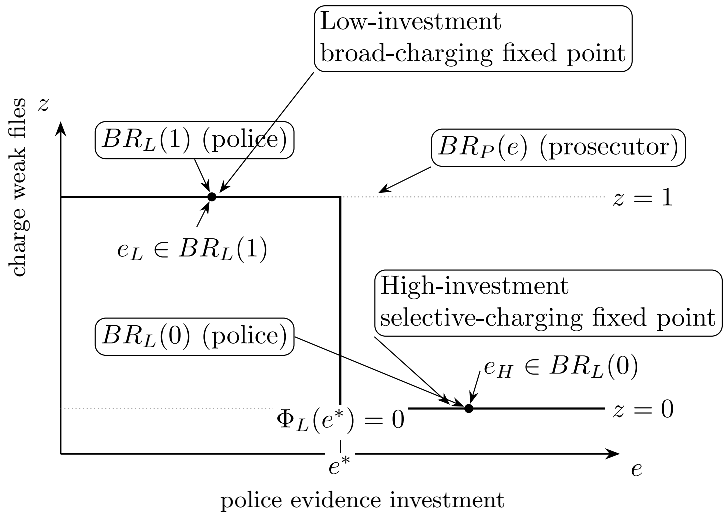

Figure 2: Best responses in the lenient continuation game. The prosecutor's best response $BR_P(e)$ is a threshold rule in police investment e. The police best response yields lower investment when weak files are charged ($z = 1$) and higher investment when weak files are declined ($z = 0$), generating two self-sustaining fixed points.

and it is here that reform can fail to shift observed behavior.

The mechanism can be read directly off Condition (3). Somewhat counterintuitively, a lenient charging standard is harder to sustain when police under-invest in effort. When police under-invest, $s = L$ is a less reliable signal of guilt, which shrinks Q_i^L toward $q(0, L)$—the conviction probability for a truly innocent defendant. The lenient regime therefore requires the prosecutor to tolerate charging weak files, but the regime itself causes weak files to be especially uninformative. A lenient prosecutorial standard thus requires *more* police effort to sustain, not less. Conversely, when police invest heavily, $s = L$ more reliably signals innocence, tilting the prosecutor toward selective (rather than broad) charging. The two best responses bend toward each other, and two self-sustaining fixed points emerge.

Figure 2 illustrates the coordination logic. When police investment is low, weak and strong files are difficult to distinguish, making broad charging privately optimal and sustaining low investment. When investment is high, weak files more reliably signal innocence, making selective charging optimal and sustaining high investment. The interaction therefore generates institutional path dependence: observed enforcement practices reflect not only current policies but the system's inherited position among multiple equilibria.

4.3 Screening, reallocation, and where reforms can "move" the system

This coordination logic interacts with the cross-category priority mechanism emphasized above. Under strict screening, evidence investment affects not only conviction probability but also the probability of being charged, which increases the marginal return to investigative effort in categories where evidence is productive. This yields the following reallocation implication (restated here to bridge from equilibrium characterization to reform):

Proposition 6 (Screening and police reallocation) *Holding police agency preferences fixed, a shift from a lenient to a strict evidentiary standard weakly increases agency priorities toward categories in which evidence investment more strongly raises the probability of charging. This holds exactly when the effort budget is slack; when it binds, it holds provided the shift in relative returns dominates the change in the budget's shadow-price term.*

Proposition 6 clarifies why reforms that credibly harden screening can change not only charging rates but also the *composition* of enforcement. But the multiplicity result implies a further caution: even if a reform changes formal standards or increases the prosecutor's stated commitment to screening, police behavior (and therefore observed outcomes) may remain similar if the reform does not shift the equilibrium selection problem in Figure 2.

Those observations in turn have direct bearings on empirical analyses of prosecutorial reform and its effects on the criminal justice system. For example, Agan et al. (2025) find that the election of reform prosecutors leads to reduced charging and conviction rates in misdemeanor cases without a similar effect in felony cases. Mitchell et al. (2022) document evidence that selecting a reform prosecutor leads to reduced racial disparities in who is charged with a crime. Further, this finding has immediate implications for the nascent literature that examines how police priorities and enforcement efforts respond to expectations about how suspected criminals will be treated in later stages of the criminal justice system (e.g., Goncalves and Mello Forthcoming, Soliman 2022). In all of these instances, our inferences from empirical relationships between policy reform and policing and prosecutorial decisions are complicated by the coordination dynamic that drives equilibrium selection.

4.4 Reform with and without change

First consider reform without observed change. In this situation, we have persistence of behavioral patterns that is due to multiplicity of equilibria that remains after reform. To state the formal result cleanly, define reform as a parameter shift that affects payoffs or institutions but not necessarily equilibrium selection.

Definition 1 (Reform as a parameter shift) *Let ω denote the relevant primitives (e.g., $a, \lambda, \rho, \Gamma, \{\alpha_i, \beta_i\}$, the effort budget $\bar{B}$, and the reallocation cost C). A* reform *is a shift $\omega \to \omega'$. Let $\mathcal{E}(\omega)$ denote the set of PBE under primitives ω.*

Under this definition, reform might be thought of as any change in the players' preferences or the technology that maps police effort into case files. Recall from above that the prosecutor's choice of charging standard is generically unique, because the model parameters determine whether $V_P(\text{strict}) > V_P(\text{lenient})$. Thus, reform in the sense of a formal change in policy requires a change in these primitives.

Proposition 7 (Reform may not change observed outcomes under equilibrium persistence) *Fix a category i and suppose that under ω the lenient continuation game sustains two equilibria: a low-investment/broad-charging equilibrium $(z, e) = (1, e_L)$ and a high-investment/selective-charging equilibrium $(z, e) = (0, e_H)$ with $e_H > e_L$.*

Consider a reform $\omega \to \omega'$ such that:

(i) both fixed points remain PBE-consistent under ω' (i.e., both remain in $\mathcal{E}(\omega')$), and

(ii) post-reform behavior remains coordinated on the pre-reform fixed point (equilibrium persistence).

Then the reform need not meaningfully change observed aggregate outcomes (e.g., charging rates, conviction rates, declination rates), even though primitives have shifted.

Proposition 7 captures the sense in which institutional practices can persist in the face of formal change. The mechanism is not merely "inertia"; it is that best responses remain mutually consistent at the pre-reform fixed point. In the low-investment equilibrium, prosecutors face an inherited evidentiary environment in which weak files do not reliably signal innocence, making broad charging privately rational; police, observing broad charging, have little reason to invest in diagnostic evidence. A reform that changes office policy statements or marginally shifts payoffs can fail to move the system if it does not eliminate the old fixed point or create a credible transition that changes expectations.

Next, consider when reform *must* bring about behavioral change. The model identifies when reforms should have large, discontinuous effects: when they push the system out of the multiplicity region by eliminating one of the fixed points in Figure 2.

Proposition 8 (Reforms that eliminate a fixed point generate nonmarginal change) *Fix i and suppose multiplicity obtains under ω as in Proposition 7. If a reform $\omega \to \omega'$ shifts primitives so that, in the post-reform continuation game, either*

$$\Phi'_L(\max BR'_L(0)) > 0 \quad or \quad \Phi'_L(\min BR'_L(1)) < 0,$$

then the lenient continuation game under ω' admits a unique equilibrium, and equilibrium behavior (and thus aggregate outcomes) must change relative to at least one pre-reform equilibrium.

Proposition 8 clarifies the practical difference between reforms that merely *tilt* incentives and reforms that *change the structure of the coordination problem.* Increasing courtroom scrutiny (sensitivity of q to the state, θ), increasing the electoral penalty for failed prosecutions through Γ, raising the costs a or λ, or improving the diagnostic productivity of evidence production (steepening $\alpha'_i(e)-\beta'_i(e)$) can each shift $\Phi_L(e)$. But unless the shift is large enough to remove one intersection in Figure 2, observed outcomes may remain close to pre-reform levels due to persistence.

4.5 Reinterpreting existing empirical findings

The distinctions drawn above are not only a guide for new research designs. They also change how we should read evidence already in the literature. Commonly observed outcomes, such as charging rates and conviction rates and their responses to reform, are equilibrium objects jointly determined by prosecutorial screening and police behavior. They therefore do not map cleanly onto prosecutorial policy alone. Each reflects both the selection of cases into prosecution and the distribution of case quality induced by police priorities and evidence investment. I develop three reinterpretations of existing findings and then state the observable signatures that distinguish regimes.

Selection and conditional conviction. The model yields a prediction that separates it from the standard alternative in which case quality is exogenous. A reform that tightens evidentiary screening should reduce charging rates without a corresponding decline in the conviction rate conditional on charge, because stricter screening draws charged cases from a stronger pool. A model in which case quality is fixed predicts instead that tighter screening lowers both the charging rate and the conditional conviction rate. The two accounts are distinguishable with data that many reform evaluations already assemble. The informative quantity is not the average effect on charging but whether the conditional conviction rate moves with it.

Heterogeneity across offense categories. Estimated reform effects often differ sharply across offense categories (e.g., Agan, Doleac and Harvey 2023, Agan et al. 2025, Mitchell et al. 2022).The category-dependent regime (Proposition 3) generates this pattern endogenously. When categories differ in baseline convictability, courtroom scrutiny, or the productivity of evidence investment, screening and police effort concentrate in some domains and not others, so the same formal reform produces different effects across offense types. What the empirical literature records as heterogeneity to be documented, the model predicts as an equilibrium signature. The model also says which domains should show the larger response: those in which evidence investment more strongly raises the probability of charging.

Divergent reform outcomes. The contrast between Philadelphia and San Francisco with which this paper opened is, on the standard reading, a story of one reform succeeding and another failing. The model offers a different reading. Broadly similar formal reforms can yield divergent system-level consequences because jurisdictions begin in different equilibria of the coordination game (Propositions 4 and 7). This reading carries a testable implication that the success-or-failure reading does not. Pre-reform enforcement patterns, and in particular the pre-reform sensitivity of charging to evidentiary strength, should predict the magnitude of the post-reform response. Work examining whether police responded endogenously to Philadelphia's reforms (Ouss and Stevenson 2023) is consistent with the mechanism, though it was not designed to test it.

The form of the police response. Recent empirical work increasingly treats the endogenous response of police to prosecutorial policy as a threat to identification and devotes analyses to detecting or ruling it out (Agan, Doleac and Harvey 2023, Ouss and Stevenson 2023). The model treats that response as a structural feature rather than a confound, and it specifies the form the response should take. Police should reallocate enforcement across offense categories toward domains where evidence is productive, and they should adjust evidence investment within categories in the direction of the prevailing screening standard (Proposition 6). This connects to the nascent literature on how police effort responds to expected downstream treatment (e.g., Goncalves and Mello Forthcoming, Soliman 2022). It also replaces an open-ended robustness question, whether police responded, with specific predictions about how they respond.

These reinterpretations share a common structure. Equilibrium regimes differ not only in average outcomes but in the structure of enforcement: in how charging responds to evidentiary strength, how police effort is allocated across categories, and how informative prosecutorial decisions are about underlying legal merit. These patterns are stable within equilibria but differ systematically across them, which yields observable *regime signatures*. Implication 1 summarizes these distinguishing features.

Implication 1 (Observable regime signatures) *Across equilibrium regimes, the model predicts:*

1. *The unscreened regime produces* high-throughput enforcement*: high charging rates, weak sensitivity of charging to evidentiary strength, low evidence quality, and case mixes tilted toward high-volume categories.*

2. *The screened regime produces* evidence-driven enforcement*: lower charging rates, strong sensitivity to evidence, higher evidence quality, and a shift toward categories with high investigative productivity.*

3. *The category-dependent regime produces* domain-differentiated enforcement*: heterogeneous charging thresholds across categories, with evidence quality improving only in domains subject to effective prosecutorial screening.*

These signatures provide a basis for identifying which equilibrium a jurisdiction occupies, and they move measurement away from headline rates. The informative objects are the sensitivity of charging to evidentiary strength and the allocation of enforcement across categories. Both can be proxied with data that are increasingly available, such as body-camera-compliance and corroboration indicators, case-file completeness, recorded dismissal reasons, and the composition of arrests across offense types. A design that tracks these quantities alongside charging and conviction outcomes can identify the charging *function*, meaning how disposition responds to case strength, rather than its average level alone.

This diagnostic also disciplines the interpretation of null results. A genuine reform can leave aggregate charging and conviction rates unchanged when the system remains coordinated on a pre-reform equilibrium (Proposition 7). Under equilibrium persistence, then, a null effect is a prediction of the model rather than evidence that prosecutorial discretion is inconsequential. The regime signatures are what distinguish persistence from irrelevance. Under persistence, the structure of enforcement is preserved along with the aggregates. Under irrelevance, there is no such structure to preserve.

Two caveats bound these claims. First, this is reinterpretation rather than identification. The model is not estimated from the studies discussed above, and consistency with its mechanism does not establish it against alternatives. Some regime signatures are observationally equivalent to explanations rooted in preferences or resources, and distinguishing them requires the measures of evidence sensitivity described above. Second, the model's sharpest predictions concern the sensitivity of charging to case strength. Where that sensitivity cannot be measured, the model speaks to the composition and direction of enforcement change but not to its precise magnitude. With those qualifications, the model predicts that pre-reform enforcement patterns should predict post-reform responses. Jurisdictions should remain near their initial equilibrium unless reforms shift both prosecutorial screening and police incentives. Models without strategic complementarity, by contrast, predict more immediate and proportional responses to policy changes.

5 Discussion and Conclusion

Taken together, the analysis here helps recast our understanding of the politics of criminal justice by highlighting the implications of the coordination game that arises because neither prosecutors nor police act independently in shaping outcomes.

Strategic interdependence and institutional governance. The core implication of the model is that prosecution and policing are jointly determined equilibrium outcomes rather than sequential stages of enforcement. Prosecutorial standards and police evidence production are strategic complements: each actor's optimal behavior depends on expectations about the other's response. As a result, formal authority over charging does not translate mechanically into control over enforcement outcomes.

This interdependence has three implications. First, it decouples authority from influence. Prosecutors set charging rules, but police determine the informational environment in which those rules operate. Screening standards shape police incentives to invest in evidence, and police investment shapes the posterior beliefs that make screening viable. Policy is therefore co-produced through equilibrium adjustment rather than imposed unilaterally. While past research has highlighted the role that police play in shaping their own internal policies (e.g., Eckhouse 2022, Mummolo 2018), in lobbying politicians for control and independence (e.g., Cheng 2024), and in setting crime-fighting priorities in the first instance (e.g., Goncalves and Mello Forthcoming), this analysis shows subtler, systemic ways in which neither police nor prosecutors can shape outcomes unilaterally.

Second, the complementarity generates regime behavior rather than marginal change: identical formal standards can sustain qualitatively distinct enforcement environments depending on how police and prosecutors coordinate, with broad implications for policy evaluation and studies of reform effects on system performance and individual welfare (e.g., Agan et al. 2025, Bandyopadhyay and McCannon 2014, Dobbie, Goldin and Yang 2018).

Third, strategic interdependence creates institutional path dependence: reforms that shift payoffs but leave the coordination problem intact may produce little observable change, so stability in enforcement patterns can reflect equilibrium persistence rather than bureaucratic inertia (cf. Sklansky 2018)—in this sense, *reforms might fail without failing.* More broadly, the analysis reframes prosecutorial reform as a problem of institutional governance in which delegation operates through endogenous information production and control depends on how oversight rules reshape agent incentives upstream. These dynamics are likely to arise in other policy domains where formally accountable principals rely on semi-autonomous agents to generate the information on which policy decisions depend.

Prosecutorial reform is fundamentally a coordination problem. Police resistance to prosecutorial reform is often attributed to differences in preferences about how to treat suspected criminals or tastes for punitiveness. The model highlights a subtler reason: when prosecutors demand stronger evidence, police are required to engage in costly effort to build stronger files, and even selectively strict standards endogenously shift both the amount of police effort and its allocation across areas of law enforcement.

Future theoretical and empirical research. While the model highlights an underappreciated strategic interaction that affects criminal justice reform, there are important avenues that are beyond the scope of this paper. Perhaps most important, we know that police are politically active and interested actors (e.g., Cheng 2024). A natural next step would be to extend the framework here to incorporate expectations about who might become the prosecutor in the future if the current prosecutor loses reelection. In the examples of both Krasner and Boudin at the opening of this paper, police organizations were active in election and recall efforts, seeking to shape voter choices. We might also imagine that potential criminals anticipate the coordination dynamics between police and prosecutors, shifting their behavior and thereby affecting the mix of potential cases for arrest and prosecution.

Finally, the analysis carries a clear agenda for empirical work, developed in detail in the section on reinterpreting existing findings above; three points bear emphasis here. Empirical work should move beyond average treatment effects of reform to study how the *sensitivity* of charging to evidentiary strength changes, since regimes can differ in that sensitivity while leaving aggregate charging and conviction rates largely unchanged. Analyses should treat policing as endogenous to prosecutorial policy, jointly tracking arrest composition, investigative effort, and dismissal patterns alongside charging and conviction outcomes. And because jurisdictions may remain coordinated on inherited equilibria, null or modest reform effects should not be read as evidence that prosecutorial discretion is inconsequential; the central challenge is identifying the equilibrium in which effects are realized, not merely estimating an average effect.

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Appendix

A-1 Supplemental Results

Proof of Proposition 1 Suppose conditions (a) and (b) hold. Consider the strategy profile in which the prosecutor commits to $\sigma =$ lenient and charges in all contingencies, $x(i, H) = x(i, L) = 1$ for both i; the police agency chooses priorities r and investments (e_A, e_B) to maximize $\mathbb{E}[U_L]$ given this charging rule. On-path beliefs are Bayesian:

$$\mu_i(s; \text{lenient}) \;=\; \frac{\pi_i \; \Pr(s \mid \theta = 1, i, e_i^*)}{\pi_i \; \Pr(s \mid \theta = 1, i, e_i^*) + (1 - \pi_i) \; \Pr(s \mid \theta = 0, i, e_i^*)},$$

evaluated at the equilibrium investments $\{e_i^*\}$ characterized below.
Charging stage. On strong files, $\Phi(i, H; \text{lenient}) > 0$ by maintained assumption, so $x(i, H) = 1$ is sequentially rational. On weak files, condition (a) gives $\Phi(i, L; \text{lenient}) \geq 0$, so $x(i, L) = 1$ is (weakly) sequentially rational.
Police best response. Officers solve the effort stage given r subject to the budget $re_A + (1-r)e_B \leq \bar{B}$. Because the prosecutor charges regardless of signal, $\Pr(x = 1 \mid i, e_i, \text{lenient}) \equiv 1$, so evidence investment affects payoffs only through the conviction margin. With $\mu \geq 0$ the multiplier on the budget, the interior officer FOC is

$$w_i \, \frac{\partial \Pr(y = 1 \mid i, e_i)}{\partial e_i} \;=\; c'(e_i) + \mu,$$

characterizing $e_i^*(\text{lenient})$ (the slack case $\mu = 0$ is the unconstrained optimum). Given the implied per-case returns $\Pi_i(\text{lenient}) = v_i + w_i \Pr(y = 1 \mid i)$, the commissioner FOC (1) characterizes $r^*(\text{lenient})$.
Policy stage. Condition (b) gives $V_P(\text{lenient}) \geq V_P(\text{strict})$, so $\sigma =$ lenient is (weakly) optimal at the policy stage given the police best response. The strategies are mutually best responses and beliefs are Bayes-consistent on path; the profile is a PBE.

■

Proof of Proposition 2 Suppose $V_P(\text{strict}) > V_P(\text{lenient})$. The construction parallels Proposition 1: the prosecutor commits to $\sigma =$ strict, σ enforces the screened charging rule $x(i, H) = 1$, $x(i, L) = 0$, the police best-respond, and on-path beliefs $\mu_i(s; \text{strict})$ are Bayesian at the equilibrium investments $\{e_i^*\}$.
The change relative to Proposition 1 is the police FOC. Under the screened rule, $\Pr(x = 1 \mid i, e_i, \text{strict}) = \Pr(s = H \mid i, e_i)$, so evidence investment now affects both the charging and conviction margins. The interior FOC becomes

$$v_i \, \frac{\partial \Pr(s = H \mid i, e_i)}{\partial e_i} \;+\; w_i \, \frac{\partial \Pr(y = 1 \mid i, e_i)}{\partial e_i} \;=\; c'(e_i) + \mu,$$

which adds the v_i term absent under unscreened charging. Because this term raises the demand for effort, the budget is more likely to bind under strict screening, so μ is weakly larger than under the lenient regime. Given the implied $\Pi_i(\text{strict}) = v_i \Pr(s = H \mid i, e_i^*) + w_i \Pr(y = 1 \mid i)$, the commissioner FOC (1) characterizes $r^*(\text{strict})$.

On strong files $\Phi(i, H; \text{strict}) > 0$ implies $x(i, H) = 1$; on weak files the strict commitment enforces $x(i, L) = 0$. The hypothesis $V_P(\text{strict}) > V_P(\text{lenient})$ makes $\sigma = \text{strict}$ optimal at the policy stage.

■

Proof of Proposition 3 Suppose $\Phi(A, L; \text{lenient}) \geq 0$, $\Phi(B, L; \text{lenient}) < 0$, and $V_P(\text{lenient}) \geq V_P(\text{strict})$. The construction combines the two preceding proofs: the prosecutor commits to $\sigma = \text{lenient}$ with the category-dependent charging rule

$$x(A, H) = x(A, L) = 1, \qquad x(B, H) = 1,\ x(B, L) = 0,$$

the police best-respond, and on-path beliefs $\mu_i(s; \text{lenient})$ are Bayesian.
Charging stage. Strong files satisfy $\Phi(i, H; \text{lenient}) > 0$, so $x(A, H) = x(B, H) = 1$. On weak files, the assumed signs of $\Phi(\cdot, L; \text{lenient})$ make $x(A, L) = 1$ and $x(B, L) = 0$ sequentially rational.
Police best response. The two categories face different marginal incentives. In category A, charging is independent of e_A, so the police FOC reduces to the conviction-only condition of Proposition 1:

$$w_A \frac{\partial \Pr(y = 1 \mid A, e_A)}{\partial e_A} \ = \ c'(e_A) + \mu.$$

In category B, charging requires $s = H$, so the FOC takes the dual-margin form of Proposition 2:

$$v_B \frac{\partial \Pr(s = H \mid B, e_B)}{\partial e_B} \ + \ w_B \frac{\partial \Pr(y = 1 \mid B, e_B)}{\partial e_B} \ = \ c'(e_B) + \mu.$$

Because both categories face the common shadow price μ, the added v_B term implies $e_B^* \geq e_A^*$, which is the investment ranking stated in the proposition. Given the implied Π_A and Π_B, the commissioner FOC (1) characterizes r^*.
Policy stage. The hypothesis $V_P(\text{lenient}) \geq V_P(\text{strict})$ makes $\sigma = \text{lenient}$ optimal.

■

Proof of Proposition 4 Fix a category i and $\sigma = \text{lenient}$. The prosecutor's best-response correspondence in z is the cutoff rule

$$BR_P(e) = \begin{cases} \{1\}, & \Phi_L(e) > 0, \\ \{0, 1\}, & \Phi_L(e) = 0, \\ \{0\}, & \Phi_L(e) < 0. \end{cases}$$

By assumption (ii), $\max BR_L(1) \leq \min BR_L(0)$.
Set $e_L = \min BR_L(1)$ and $e_H = \max BR_L(0)$. Assumption (iii) gives $\Phi_L(e_L) > 0$ and $\Phi_L(e_H) < 0$, hence $BR_P(e_L) = \{1\}$ and $BR_P(e_H) = \{0\}$. Both $(z, e) = (1, e_L)$ and $(z, e) = (0, e_H)$ are therefore mutually consistent best responses, supported by Bayesian beliefs derived from the signal distribution at the corresponding e. Assumption (i) is not needed for existence of these two fixed points but ensures Φ_L has a unique sign-crossing, ruling out further fixed points between e_L and e_H.

■

Proof of Proposition 5 Each part is a sign statement about how Φ in equation (2) and the prosecutor's value V_P depend on primitives.

Part (i). $\Phi(i, L; \text{lenient})$ is linear in b with coefficient $\Pr(y{=}1 \mid x{=}1, i, L) > 0$, linear in a with coefficient $-\Pr(y{=}0 \mid x{=}1, i, L) < 0$, linear in λ with coefficient $-(1 - \mu_i(L; \text{lenient}))q(0, L) \leq 0$, and additively separable in $\Delta_E(i, L; \text{lenient})$. The claim follows.

Part (ii). Proposition 1 requires $\Phi(i, L; \text{lenient}) \geq 0$ for both i and $V_P(\text{lenient}) \geq V_P(\text{strict})$. Part (i) gives the monotonicity of the first condition. The same primitive shifts move $V_P(\text{lenient})$ in the same direction through the prosecutor's realized payoffs on charged-and-lost and wrongful-conviction events, so the joint feasibility set inherits the stated monotonicity. The claim for Proposition 2 follows from reversing the payoff inequality.

Part (iii). The selectively screened regime requires $\Phi(A, L; \text{lenient}) \geq 0$ and $\Phi(B, L; \text{lenient}) < 0$. Under symmetry across i these coincide, so the supporting set is empty. The expansion under heterogeneity follows from continuity of Φ in $(\pi_i, \alpha_i(\cdot), \beta_i(\cdot), q(\cdot, \cdot))$.

Part (iv). Under $\sigma = \text{strict}$, the marginal return to evidence in Proposition 2 contains the term $v_i\, \partial \Pr(s{=}H \mid i, e_i)/\partial e_i$. Increases in $\alpha_i(e) - \beta_i(e)$ raise this term, and increases in $q(1, L) - q(0, L)$ raise the conviction payoff to charging strong relative to weak files. Both shifts weakly raise $V_P(\text{strict}) - V_P(\text{lenient})$, expanding the set on which screened enforcement is supported.

■

Proof of Proposition 9 Under $V_P(\text{strict}) \neq V_P(\text{lenient})$ the prosecutor has a strict preference at the policy stage, so the equilibrium value of σ is unique. Write $\Phi_L^i(e) \equiv \Phi(i, L; \text{lenient})$.

Part (i). If $V_P(\text{strict}) > V_P(\text{lenient})$, the prosecutor selects $\sigma = \text{strict}$. The strict commitment fixes the charging rule at $x(i, H) = 1$, $x(i, L) = 0$, and the police best responses (r^*, e^*) are pinned down by the FOCs in the proof of Proposition 2. On-path play is invariant across PBE.

Part (ii). If $V_P(\text{lenient}) > V_P(\text{strict})$, the prosecutor selects $\sigma = \text{lenient}$ and the equilibrium regime is determined by the lenient continuation game.

(a) If $\Phi_L^i(e) \geq 0$ for all $e \in BR_L(0) \cup BR_L(1)$ and both i, the prosecutor charges weak files everywhere on the relevant range, so $z = 1$ in both categories and the police best-respond on $BR_L(1)$. The unique fixed point is the unscreened regime of Proposition 1.

(b) If $\Phi_L^A(e) \geq 0$ and $\Phi_L^B(e) < 0$ for all $e \in BR_L(0) \cup BR_L(1)$, the prosecutor's category-by-category best response is $z = 1$ in A and $z = 0$ in B at every investment level. The unique fixed point is the selectively screened regime of Proposition 3.

(c) If, in at least one category, the crossing condition (iii) of Proposition 4 holds—$\Phi_L^i(\min BR_L(1)) > 0$ and $\Phi_L^i(\max BR_L(0)) < 0$—then by Proposition 4 the continuation game admits two PBE-consistent fixed points in that category, $(1, e_L)$ and $(0, e_H)$, and multiplicity obtains within the lenient regime.

■

Proof of Proposition 6 Under $\sigma = \text{lenient}$, $\Pi_i(\text{lenient}) = v_i + w_i \Pr(y{=}1 \mid i)$. Under $\sigma = \text{strict}$, $\Pi_i(\text{strict}) = v_i \Pr(s{=}H \mid i, e_i^*) + w_i \Pr(y{=}1 \mid i)$, evaluated at the equilibrium investment e_i^* characterized in Proposition 2. The change in the priority differential induced by switching to strict screening is therefore

$$[\Pi_A(\text{strict}) - \Pi_B(\text{strict})] - [\Pi_A(\text{lenient}) - \Pi_B(\text{lenient})] = v_A[\Pr(s{=}H \mid A, e_A^*) - 1] - v_B[\Pr(s{=}H \mid B, e_B^*) - 1].$$

The right-hand side is increasing in the relative responsiveness of $\Pr(s{=}H \mid i, e_i^*)$ to e_i: the category in which evidence investment more strongly raises the probability of charging contributes a larger (less negative) term. Through the commissioner FOC (1) and the convexity of C, this shifts r^* weakly toward that category, holding fixed the budget's shadow-price term $\mu(e_A - e_B)$. When the budget binds, that term adds a countervailing force if the favored category also draws more effort; the net reallocation is toward the more charge-productive category provided the shift in relative returns dominates the change in the shadow-price term (and exactly so when the budget is slack).

■

Proof of Proposition 7 Aggregate outcomes (charging rates, conviction rates, declination rates) are determined by the equilibrium strategies (z, e) together with the components of ω that enter the signal and conviction technologies, $\{\alpha_i, \beta_i, q\}$. Let (z^*, e^*) denote the pre-reform fixed point on which play is coordinated. By hypothesis, $(z^*, e^*) \in \mathcal{E}(\omega')$ and post-reform play remains at (z^*, e^*), so equilibrium strategies are unchanged across the reform. Reforms whose primitive shift $\omega \to \omega'$ leaves $\{\alpha_i, \beta_i, q\}$ unchanged—i.e., shifts in payoff parameters such as a, λ, ρ, or Γ—therefore leave aggregate outcomes exactly unchanged. More generally, any reform that perturbs $\{\alpha_i, \beta_i, q\}$ only locally at (z^*, e^*) produces a correspondingly small change in aggregate outcomes. Hence the reform need not meaningfully change observed outcomes, even though primitives have shifted.

■

Proof of Proposition 8 Fix i. Multiplicity under ω requires both

$$\Phi_L(\min BR_L(1)) > 0 \quad \text{and} \quad \Phi_L(\max BR_L(0)) < 0$$

by Proposition 4. Reform reverses one of these inequalities.
Suppose first that $\Phi'_L(\max BR'_L(0)) > 0$. Since Φ'_L is decreasing in e by assumption (i), $\Phi'_L(e) \geq \Phi'_L(\max BR'_L(0)) > 0$ for all $e \in BR'_L(0)$, so $BR'_P(e) = \{1\}$ for every $e \in BR'_L(0)$. No element of $BR'_L(0)$ can therefore support $z = 0$, and the selective-charging fixed point is eliminated.
Symmetrically, if $\Phi'_L(\min BR'_L(1)) < 0$, then $\Phi'_L(e) < 0$ for all $e \in BR'_L(1)$, so $BR'_P(e) = \{0\}$ throughout $BR'_L(1)$ and the broad-charging fixed point is eliminated.
In either case only one branch survives, so the lenient continuation game has a unique equilibrium under ω'. Since both pre-reform fixed points $(1, e_L)$ and $(0, e_H)$ were equilibria under ω but only one branch remains, the post-reform equilibrium cannot coincide with both pre-reform fixed points; equilibrium behavior must change relative to at least one of them.

■

A-2 Modeling choices

There are several choices in the model set-up that bear discussion. First, I model the prosecutor's evidentiary standard as a discrete, publicly observed choice. Moreover, σ is not a costless announcement or aspirational guideline. Rather, it is a commitment device that constrains the prosecutor's subsequent charging decisions. Formally, the choice of σ restricts the set of admissible charging rules available to the prosecutor after observing (i, s). This formulation captures the idea that internal screening rules—such as corroboration requirements, mandatory body-camera compliance, or supervisory approval thresholds—shape which cases may be filed, even though prosecutors retain discretion within those bounds.

The binary specification deserves a more explicit defense, because one could instead model σ as a continuous evidentiary threshold. I make the discrete choice for three reasons. First, institutional realism: real prosecutorial offices do not publish continuous evidentiary thresholds. They adopt coarse, enumerable policies—case-type declination rules, mandatory corroboration, bright-line filing criteria tied to specific evidentiary categories—that approximate a binary distinction between stringent and permissive regimes far better than a continuum (cf. Barkow 2019, Davis 2007). The two case studies motivating the paper—the Krasner and Boudin reforms—are routinely described by both reformers and critics in categorical terms, as shifts between qualitatively different charging postures. Second, commitment logic: a binary, publicly observed standard is easier to make credible than a continuous threshold. Precisely because it is coarse, it is verifiable: staff, journalists, and police can identify whether the office is observing the rule, which is what gives the standard its commitment bite in the first place. A continuous threshold would have to be either implemented as a fine-grained internal rule that is difficult to monitor, or coarsened in practice into a binary one, returning to the present formulation. Third, analytic tractability and focus: the core mechanism of the paper—the strategic complementarity between police evidence investment and prosecutorial screening, and the multiplicity it generates—does not depend on having more than two values of σ. Adding a continuum would reintroduce notational complexity without changing the qualitative results characterized below. The binary formulation therefore isolates the substantive interaction of interest while maintaining transparency about which conclusions depend on which modeling assumptions. In practice, internal screening rules are more nuanced, but the binary formalization captures the core distinction between offices that require extensive corroboration before filing and those that do not.

Second, I assume that the prosecutor does not directly observe the police agency's evidence investment e_i. Instead, the prosecutor conditions her choice on the police file, s (i.e., the noisy signal generated in part by effort), and on her own standard σ, which pins down the equilibrium investment level. This captures the realistic feature that a prosecutor can evaluate the strength of the case file she receives but cannot perfectly monitor the effort that went into producing it.

Third, I simplify the courtroom environment by representing it with the exogenous function $q()$. In typical local criminal courts, prosecutors play extremely influential roles, with repeated, regular appearances before judges who often defer to them. Endogenizing judicial behavior is an important question, but one I set aside in order to focus analytic attention on the police-prosecutor interaction.

Fourth, I treat the police agency as two roles rather than one. A commissioner sets enforcement priorities, and line officers choose investigative effort on the cases they draw. Officers move after the commissioner and share a fixed investigative budget, so the effort available per case depends on how the commissioner allocates cases across categories. This keeps the agency's objective unified while making the commissioner's priority choice depend on the effort she can anticipate from officers, which is where the interaction between allocation and effort resides. The budget is what couples the two choices; a separate convex reallocation cost keeps the priority choice interior. The two frictions are distinct, and the model reduces to a single-stage agency problem when the budget does not bind.

Fifth, the charging margin is best understood as applying to contested cases. The overwhelming majority of criminal cases resolve by plea, and an evidentiary standard bites most sharply where the prospect of trial disciplines charging. I therefore read σ and the weak-file margin as governing the contested cases in which trial viability is a live consideration. Cases that would resolve by plea regardless of the standard enter as a background that shifts the levels of aggregate outcomes but not the comparative statics of interest.

Sixth, I treat evidence as produced by police before a case reaches the prosecutor, abstracting from prosecutor-initiated investigation. Many offices allow prosecutors to request additional evidence or to deploy their own investigators. Adding that stage would not alter the strategic core: such requests are themselves a function of the anticipated charging standard, so they reinforce rather than dissolve the complementarity between screening and evidence production. Modeling them explicitly would complicate the sequence of play while leaving the equilibrium logic intact.

A-3 Comparative statics: parameter channels and regime unique-

ness

This appendix unpacks the three channels behind Proposition 5 and states the uniqueness structure of the equilibrium regimes referenced in the main text.

Electoral incentives to charge. The electoral marginal incentive $\Delta_E(i, s; \sigma)$ shapes the prosecutor's willingness to file weak cases. When voters reward visible enforcement and weakly penalize acquittals or dismissals, $\Delta_E(i, L; \sigma)$ is positive even for low-quality files, pushing the system toward unscreened charging. When electoral rewards are tied to competence or successful outcomes, charging weak cases becomes costly, creating scope for selective screening. When prosecutors are disproportionately rewarded for charging lots of cases rather than for the distribution of case outcomes, they are less likely to impose strict standards, a result that follows immediately from part (i) of Proposition 5. As we will see below, that incentive in turn shapes the quality of policing in nuanced ways.

Costs of failed or unjustified prosecutions. The parameters a and λ raise the effective charging threshold by increasing the expected cost of acquittals and wrongful convictions. High values discipline charging behavior even under electoral pressure to be tough, supporting screened charging equilibria. When both costs are small, electoral incentives dominate, making unscreened charging more likely.[3]

Evidence technology, courts, and heterogeneity. The diagnostic productivity of evidence investment and courtroom scrutiny $q(\cdot, \cdot)$ determine how sharply weak files differ from strong ones. When investment is productive and scrutiny is meaningful, charging decisions become sensitive to evidence quality; when not, evidentiary distinctions carry little payoff. Heterogeneity across categories generates selectively screened equilibria, with discipline applied only in some domains—the content of parts (iii) and (iv) of Proposition 5.

When is each regime unique? Propositions 1–3 give sufficient conditions for the *existence* of each regime. Because these sufficient conditions can sometimes be satisfied simultaneously with

[3]Notice that a appears on both sides of Condition (3) but is multiplied by a probability on the left-hand side but not on the right-hand side. Thus, increasing a makes the condition weakly harder to satisfy, consistent with part (i) of Proposition 5.

the conditions of Proposition 4, it is useful to state explicitly when each regime is the unique PBE regime of the game. The following proposition characterizes this uniqueness structure.

Proposition 9 (Uniqueness of the equilibrium regime) *Suppose the primitives of the game satisfy the generic condition $V_P(strict) \neq V_P(lenient)$.*

(i) If $V_P(strict) > V_P(lenient)$, the screened regime of Proposition 2 is the unique PBE regime, and on-path play is invariant across PBE.

(ii) If $V_P(lenient) > V_P(strict)$, the regime played in equilibrium is determined in the lenient continuation game:

(a) if $\Phi(i, L; lenient) \geq 0$ for all $i \in \{A, B\}$ at every $e \in BR_L(0) \cup BR_L(1)$, the unscreened regime of Proposition 1 is the unique PBE regime;

(b) if, without loss of generality, $\Phi(A, L; lenient) \geq 0$ and $\Phi(B, L; lenient) < 0$ at every $e \in BR_L(0) \cup BR_L(1)$, the selectively screened regime of Proposition 3 is the unique PBE regime;

(c) if, in at least one category, the crossing condition of Proposition 4 holds, both the unscreened (or selective) and the high-investment/selective-charging fixed points are PBE-consistent, and multiplicity obtains within *the lenient regime.*

Proposition 9 sharpens the relationship between the first-mover policy choice and the continuation game. The prosecutor's choice of σ is generically unique—the knife-edge $V_P(\text{strict}) = V_P(\text{lenient})$ is a measure-zero set of primitives—so multiplicity of regimes never arises across the strict–lenient boundary. Instead, all genuine multiplicity in the model is *intra-regime*: it lives inside the lenient continuation game, where the prosecutor's residual discretion over weak files interacts with police investment to support more than one mutually consistent pair of best responses. The upshot is that the three regimes partition the primitive space into well-defined cells (cases (i), (ii.a), and (ii.b) of Proposition 9), with a distinct cell (case (ii.c)) in which the unscreened and selective regimes coexist as equilibria. This partition motivates the discussion of equilibrium selection and reform in the remainder of the section.